

AI-Driven Design and Property Prediction Framework for Biopolymer Bone Scaffolds

Predicting how a scaffold will perform — and how it will fail as it degrades — before a single one is printed.

SIMULATED TIER

747 scaffolds · 10 architectures

FORWARD MODEL

zero learned parameters

VALIDATION

leave-one-architecture-out

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PROGRAMME

B.Tech, Artificial Intelligence & Machine Learning

PROJECT GUIDE

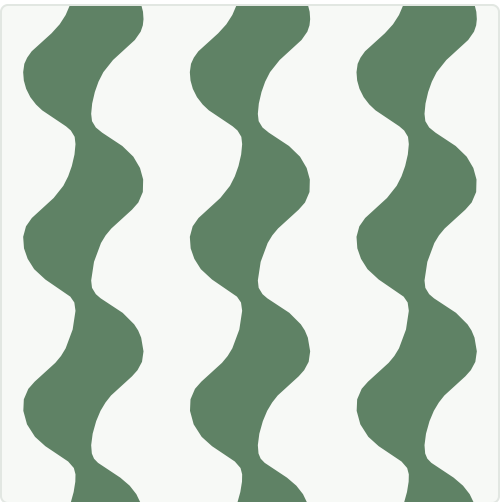
Mr. Athul Ravi

GYROID SECTION

0.35
POROSITY



0.60
POROSITY



0.85
POROSITY



Level set of the gyroid as the polymer resorbs. Solid phase shown; slices computed from the project's own TPMS generator.

What the project is, and why it matters

THE PROBLEM

Scaffold development is still print–test–repeat. Every loop costs weeks, and it only ever measures the one geometry you already built.

It also measures the wrong day. A resorbable polyester loses molecular weight, then mass — so porosity climbs and stiffness falls away exactly while the bone is still healing.



🔄 repeat — weeks per iteration, one geometry at a time

41 %

of day-one stiffness left at 12 weeks

PLGA 85:15 / β -TCP — what the loop never measures

This framework replaces the loop with a forward model: composition and architecture in, a full stiffness-versus-time trajectory out.

WHAT IT BUYS YOU — EVERY FIGURE READ FROM THE REPOSITORY

747

simulated scaffolds across 10 architectures

FDM strut lattices and four TPMS families. Porosity 0.16–0.91, every row a converged voxel-FEM solve.

metrics.json · dataset

0.73

R² on a topology never seen in training

Leave-one-architecture-out — the question a design tool has to answer, not the easy one.

metrics.json · model_ladder

0.99 → 0.53

what an honest split actually costs

Random K-fold claims R² 0.99; grouping by architecture gives 0.53. That 0.47 gap is manufactured optimism.

metrics.json · leakage.simulated

725 → 266

geometries survive the clinical gates

Pore 200–600 μm , porosity 50–90 %, strut $\geq$ 150 μm , open pores. Hard rejects, not penalty weights.

metrics.json · inverse

How the prediction works

MECHANISTIC DECODER *composition + architecture + time* → *E(t)*

1

Composition

weight fractions → volume fractions

densities, handbook

2

Halpin–Tsai

solid composite modulus E_s

Halpin 1969

3

Autocatalytic ODE

$M_n(t)$, then mass loss

Antheunis et al. 2010

4

Solubility cutoff

evolving porosity $P(t)$

critical M_n , entanglement

5

Gibson–Ashby

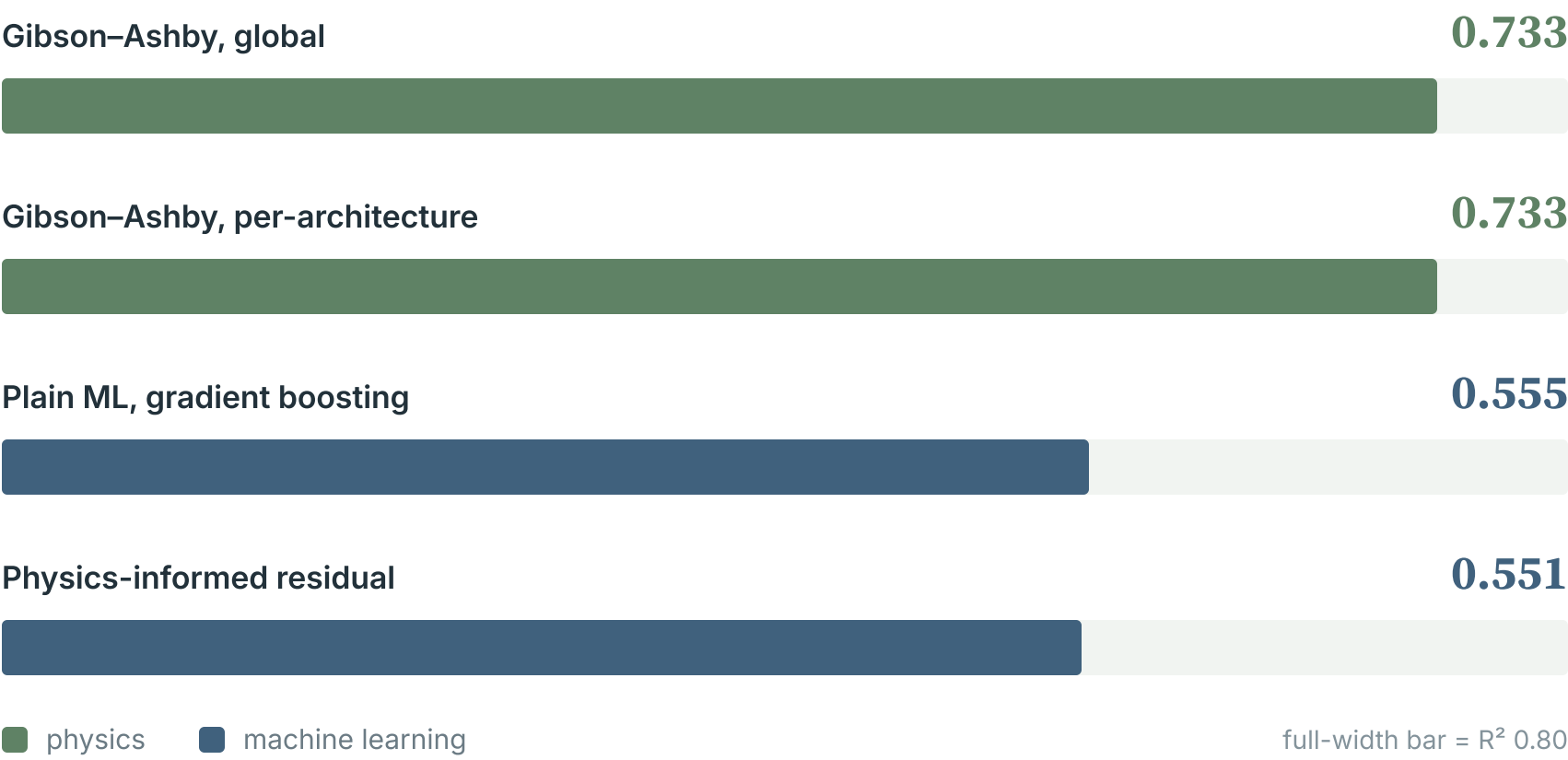
$E(t) = C \cdot E_s(t) \cdot (1 - P(t))^n$

Gibson & Ashby 1982

Zero learned parameters in this chain.

Only (n, C) are fitted — and C is re-anchored to the FEM solution at t = 0, so the trajectory starts on the modulus that was actually solved.

LEAVE-ONE-ARCHITECTURE-OUT R^2 *held-out topology* · 736 scored rows



The physics wins. The ML ladder is a benchmark, not the result.

The residual is fitted on architectures it has seen, so on a held-out topology the learned correction is not merely uninformative — it is wrong, and it is added to a prior that was already right.

The two physics bars are identical by construction: with the topology held out, the per-architecture lookup has no entry and falls back to the global fit.